



Kabul University
Computer Science Faculty
Information Technology Department

Computer Architecture

Outline of the Book

- Computer Evolution and Performance
- Computer Interconnection Structures
- Internal Memory
- External Memory
- Input/output
- Operating Systems Support
- Computer Arithmetic
- Instruction Sets
- CPU Structure and Function
- Superscalar Processors
- Control Unit Operation

Internet Resources

- Web site for book

- <http://WilliamStallings.com/COA/COA7e.html>
 - links to sites of interest
 - links to sites for courses that use the book
 - errata list for book
 - information on other books by W. Stallings
- <http://WilliamStallings.com/StudentSupport.html>
 - Math
 - How-to
 - Research resources
 - Misc

Internet Resources

- Web sites to look for

- WWW Computer Architecture Home Page
- CPU Info Center
- Processor Emporium
- ACM Special Interest Group on Computer Architecture
- IEEE Technical Committee on Computer Architecture
- Intel Technology Journal
- Manufacturer's sites
 - Intel, IBM, etc.

Course Objectives

- To learn
- Computer types
- Functional units
- Basic operational concepts
- Number representation and arithmetic operations
- Character representation
- Performance
- Historical perspective
 - How computers work, basic principles
 - How to analyze their performance
 - How computer are designed and built
 - Issues effecting modern processors (processors)

Course Motivation

- This course will be useful if you need to
 - Design/build a new computer
 - Design/build a new version of a computer
 - Improve software performance
 - Purchase a computer
 - Provide a solution with an embedded computer

Computer organization/Computer architecture

Computer organization

- Design of the **components and functional blocks** using which computer systems are built.
- For example: in computer system we have functional blocks such as processor inside processor we have many parts like ALU, Register and so on, we use adder in ALU so which kinds of adder must be used, this is the focuses area of computer organization.
- Computer architecture
 - How to integrate the computer components to achieve the desired level of **performance**.

Cont.

Computer Organization

- Often called microarchitecture (Low level)
- Physical components (Circuit design, adder, Signals, Peripheral)

Computer architecture

- Computer architecture (a bit higher level)
- Programmer view (Ex. Programmer has to be aware of which instruction set used)
- Logic(instruction set, addressing modes, data types, cache optimization)

Cont.

- By the term computer architecture it is generally mean those qualities of a system that are visible to a programmer, such as the number of bits used to represent various data types, the instruction set of the computer, techniques for addressing memory, methods used for input-output, etc.

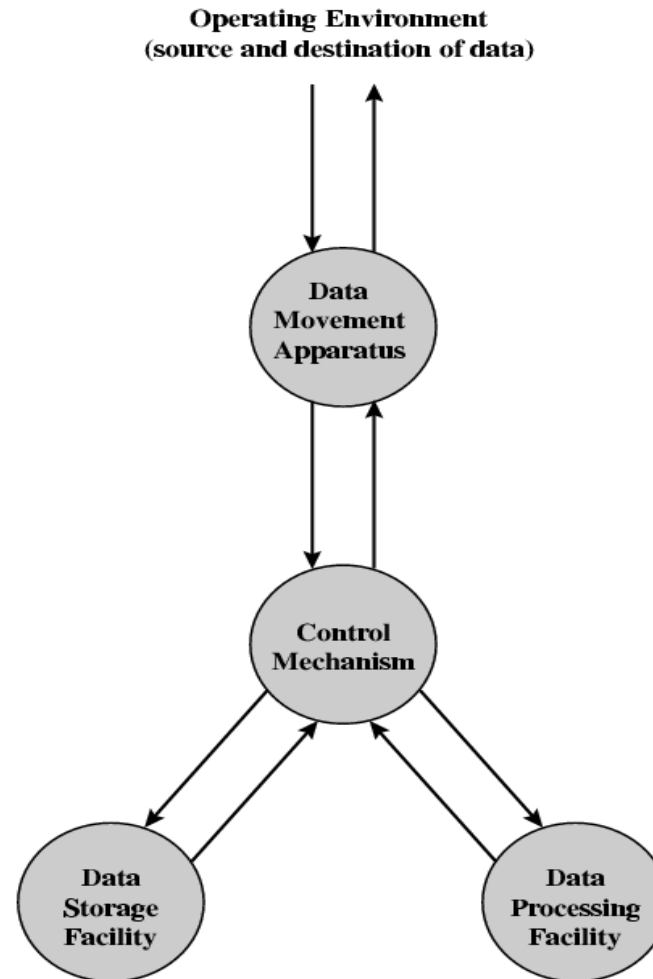
Structure & Function

- Structure is the way in which components relate to each other
- Function is the operation of individual components as part of the structure

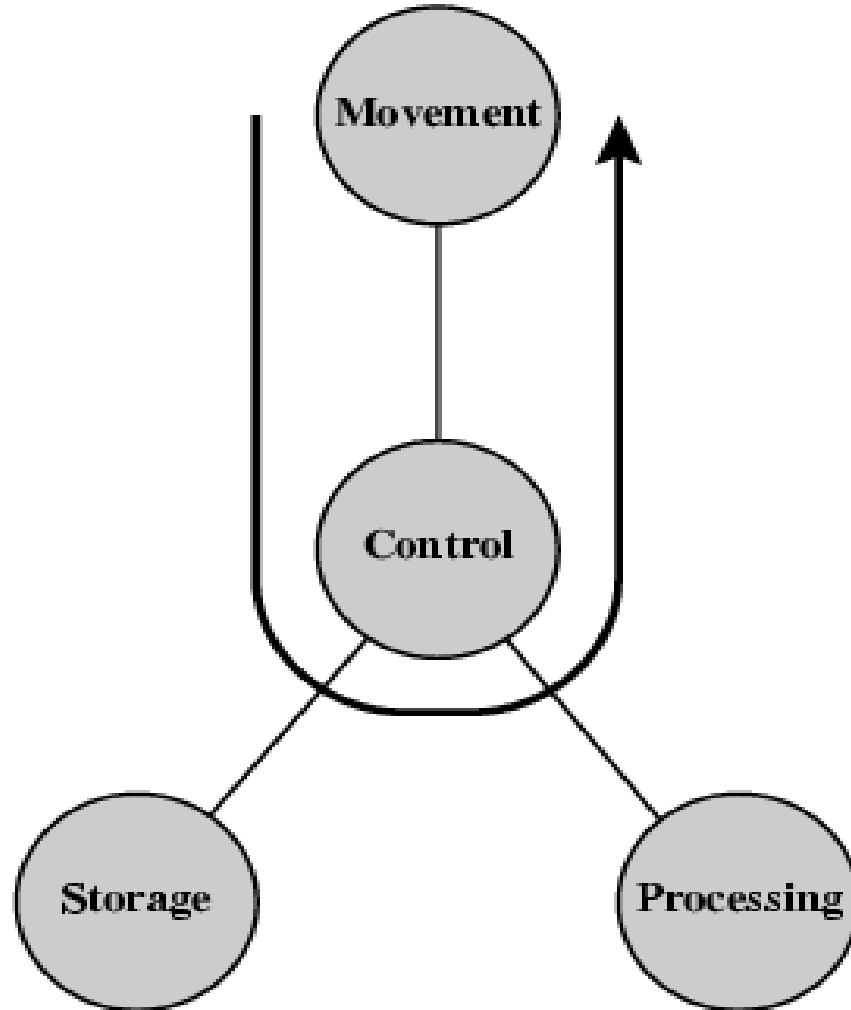
Function

- All computer functions are:
 - Data processing
 - Data storage
 - Data movement
 - Control

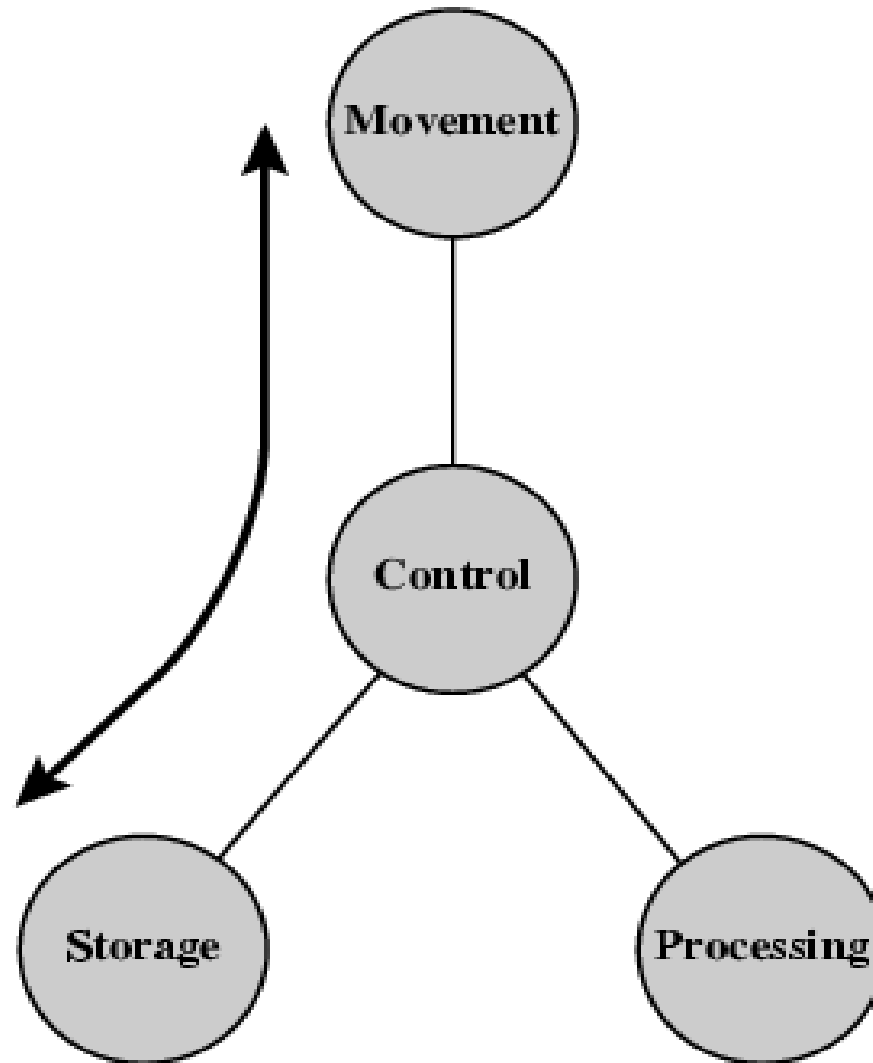
Functional View



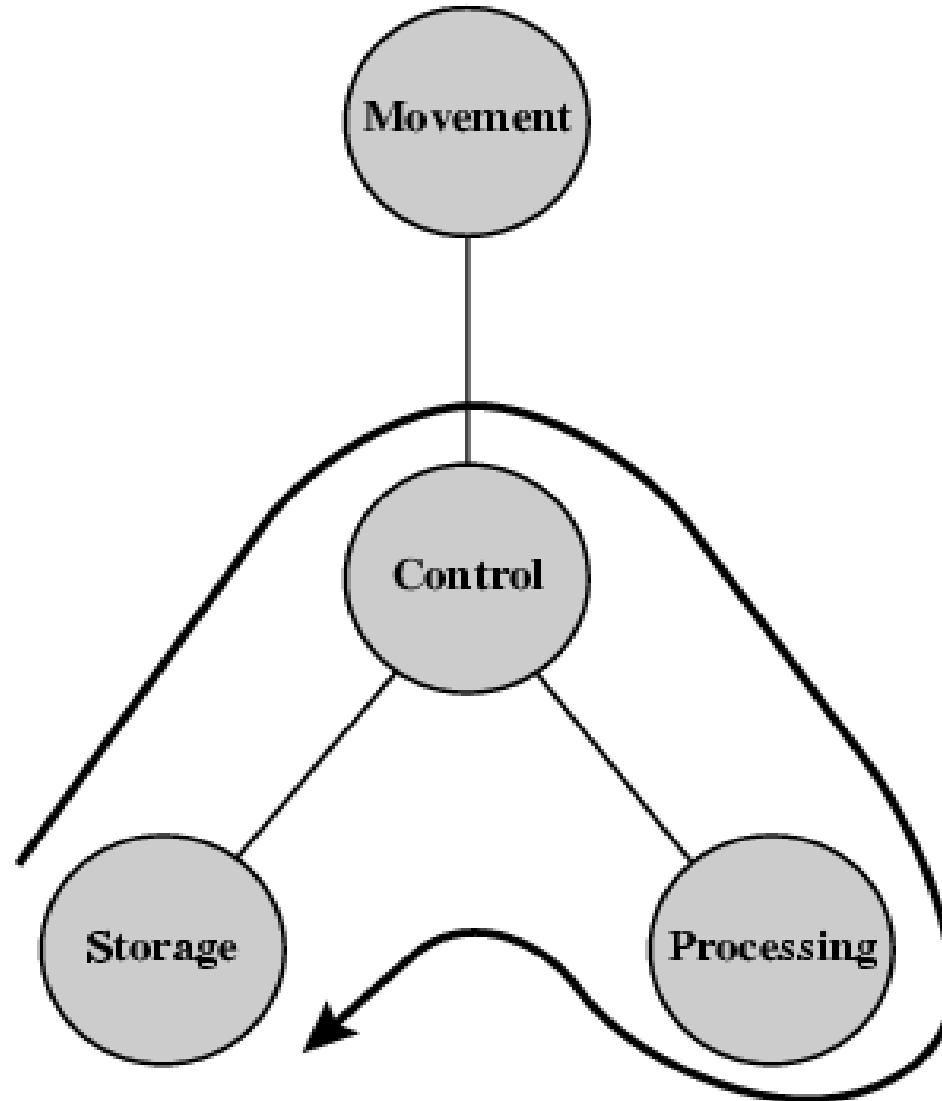
Operations (a) Data movement



Operations (b) Storage

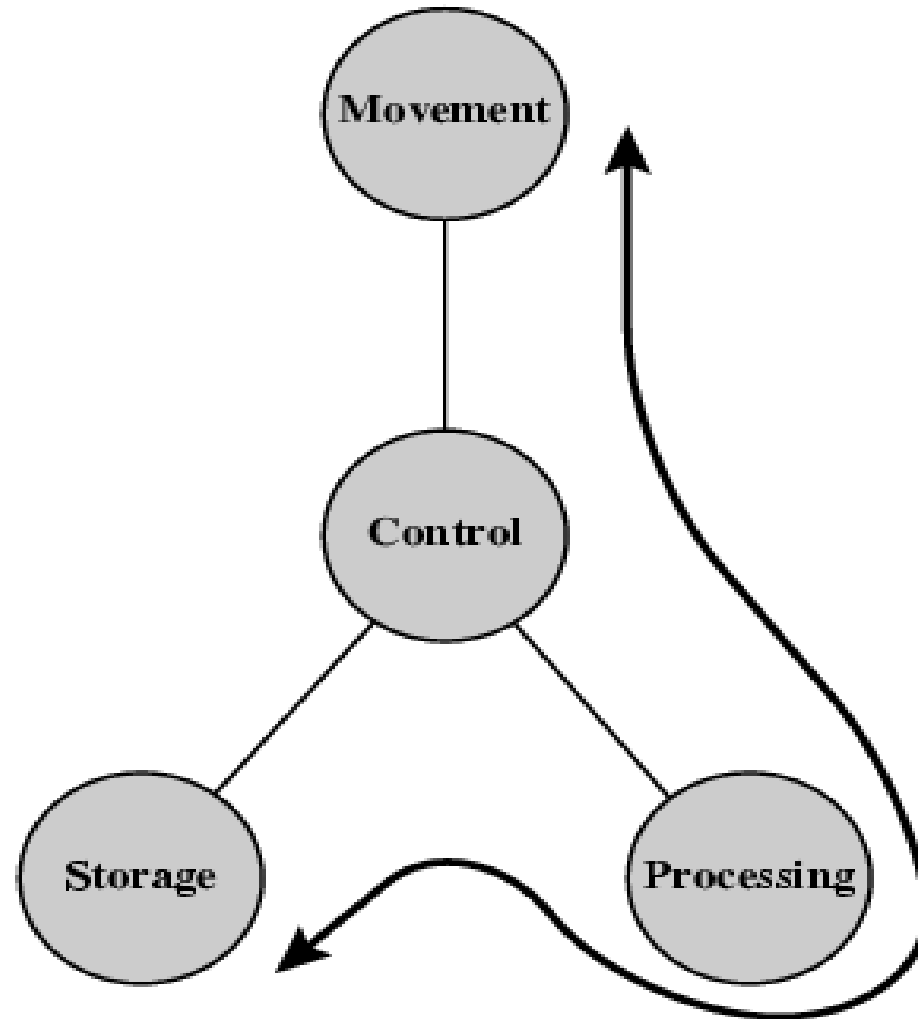


Operation (c) Processing from/to storage

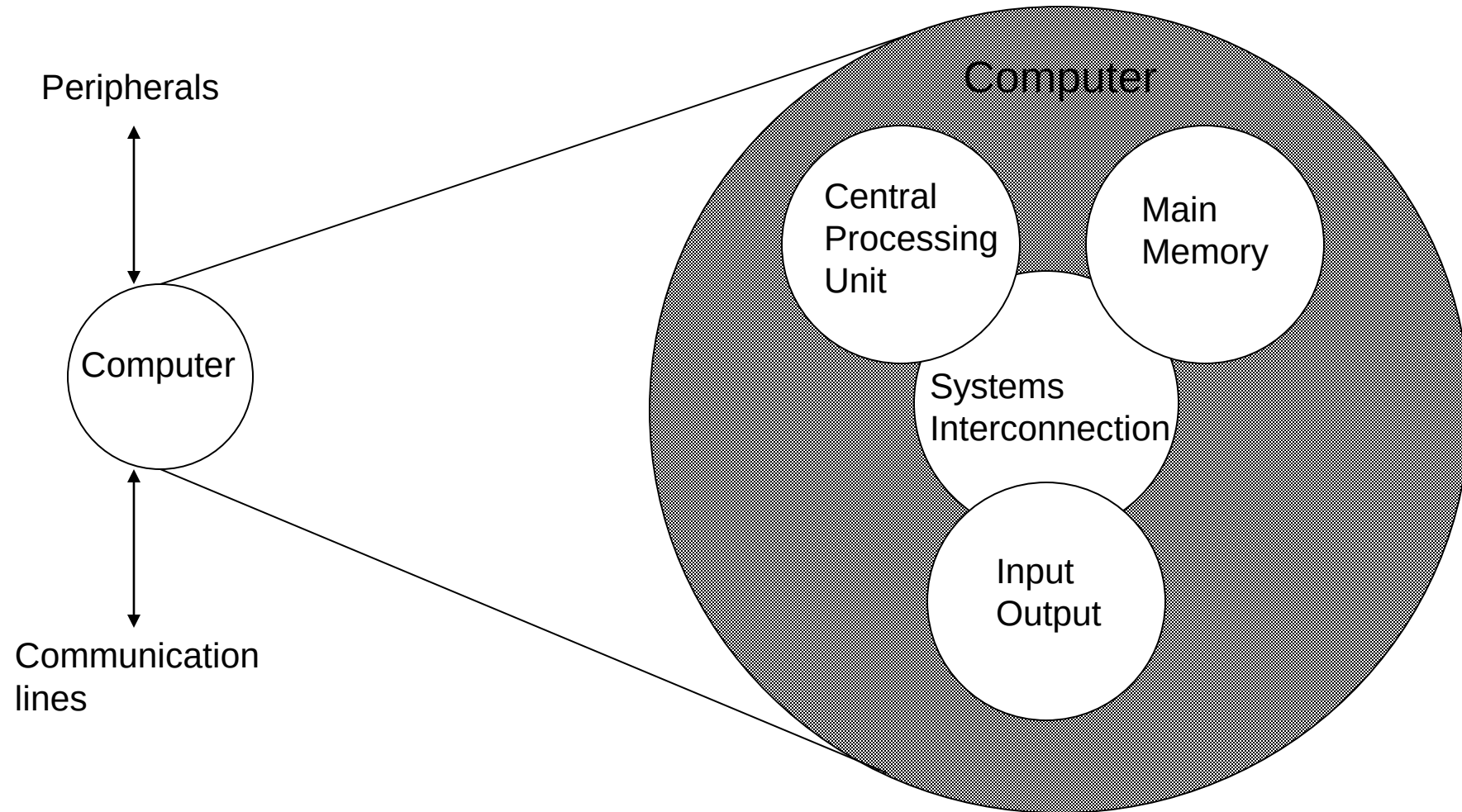


Operation (d)

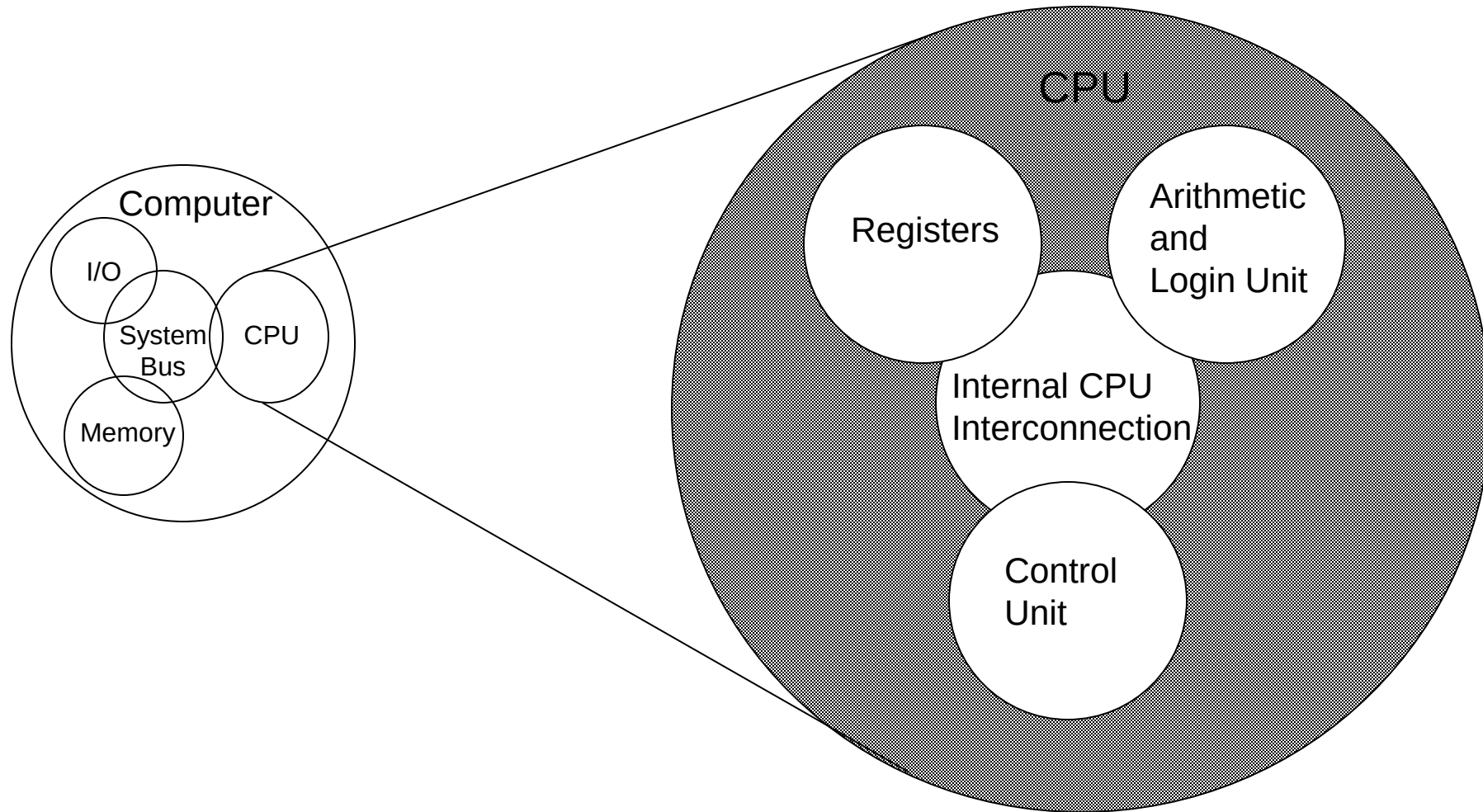
Processing from storage to I/O



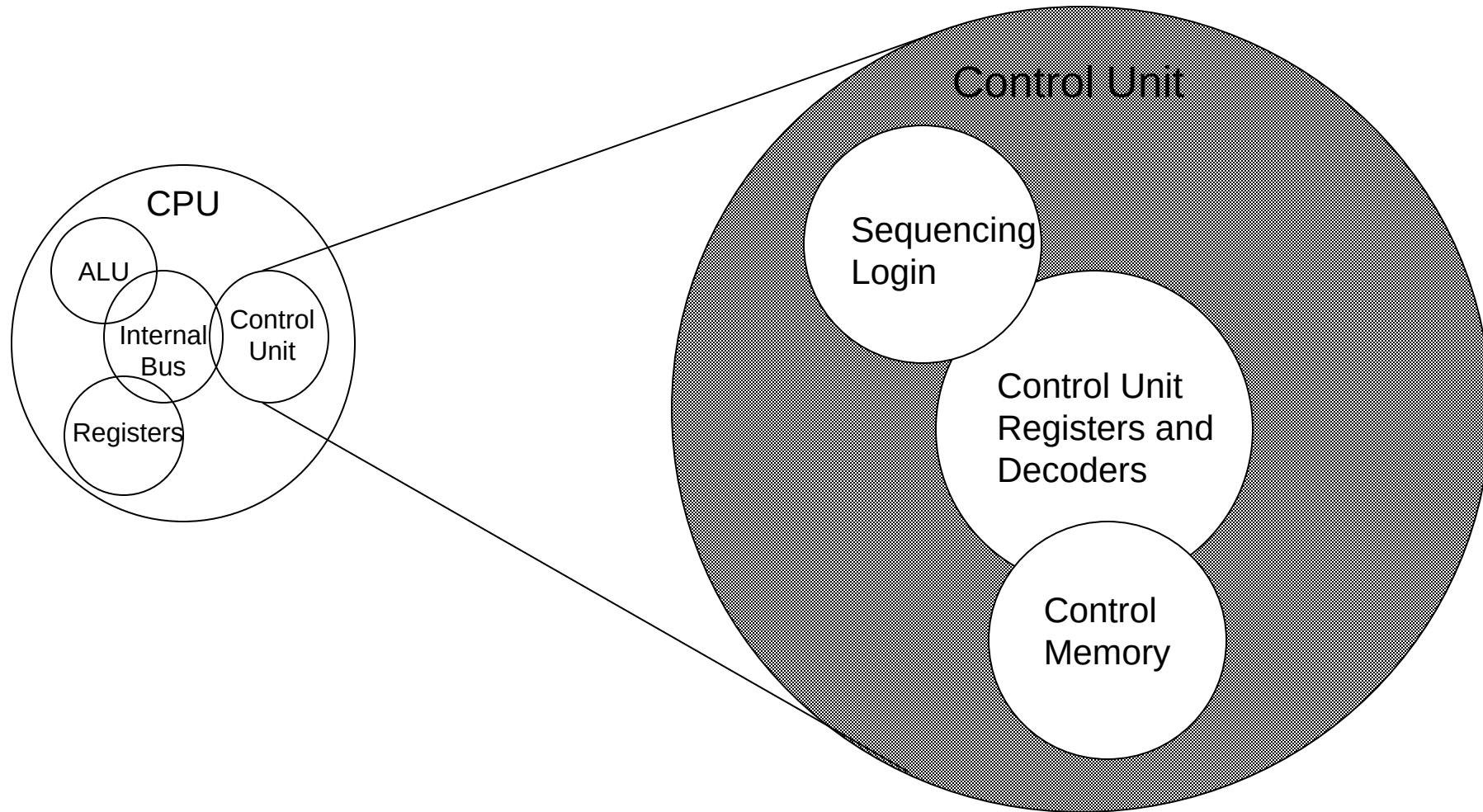
Structure - Top Level



Structure - The CPU



Structure - The Control Unit



Thanks for your attention